

Day 11 - Routing Fundamentals (pt 1)

Saturday, August 22, 2026 2:50 PM

11. ROUTING FUNDAMENTALS : PART 1

WHAT IS ROUTING ?

ROUTING is the process that routers use to determine the path that IP packets should take over a network to reach their destination.

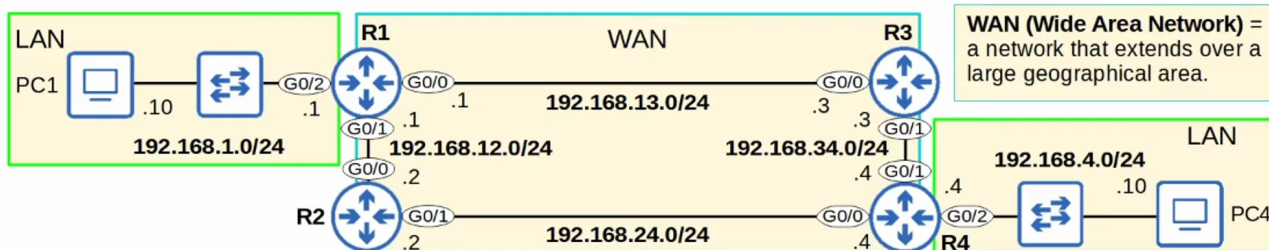
- ROUTERS store routes to all their known destinations in a ROUTING TABLE
- When ROUTERS receive PACKETS, they look in the ROUTING TABLE to find the best route to forward that packet.

There are two main routing methods (methods that routers use to learn routes):

- **DYNAMIC ROUTING** : ROUTERS use Dynamic Routing Protocols (ie: OSPF) to share routing information with each other automatically and build their routing tables.
- **STATIC ROUTING** : A network engineer / Admin manually configures routes on the router.

A ROUTE tells the ROUTER :

- to send a packet to Destination X, you should send the pack to *next-hop* Y
- or if the Destination is directly connected to the router, *send the packet directly to the destination*.
- or if the Destination is the router's own IP address, *receive the packet for yourself (don't forward it)*.



WAN (Wide Area Network) = network that extends over a large geographic area.

R1 Pre-configurations (IP Addresses)

```
R1# conf t
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.13.1 255.255.255.0
R1(config-if)# no shutdown

R1(config-if)# interface g0/1
R1(config-if)# ip address 192.168.12.1 255.255.255.0
R1(config-if)# no shutdown

R1(config-if)# interface g0/2
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

There is no need to use **exit** to return to global config mode before entering **interface g0/1**. You can use the **interface g0/1** command directly from interface config mode.

```
R1# show ip int br
Interface                               IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0                     192.168.13.1    YES manual up          up
GigabitEthernet0/1                     192.168.12.1    YES manual up          up
GigabitEthernet0/2                     192.168.1.1     YES manual up          up
GigabitEthernet0/3                     unassigned      YES NVRAM   administratively down down
```

Shop ip route

Routing Table (show ip route)

R1# show ip route Use the command **show ip route** to view the routing table.

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

The Codes legend in the output of **show ip route** lists the different protocols which routers can use to learn routes.

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

```

C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.1.0/24 is directly connected, GigabitEthernet0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet0/2
C    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.12.0/24 is directly connected, GigabitEthernet0/1
L    192.168.12.1/32 is directly connected, GigabitEthernet0/1
C    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.13.0/24 is directly connected, GigabitEthernet0/0
L    192.168.13.1/32 is directly connected, GigabitEthernet0/0
  
```

- **L - local**
→ A route to the actual IP address configured on the interface. (with a /32 netmask)
- **C - connected**
→ A route to the network the interface is connected to. (with the actual netmask configured on the interface)

When you configure an IP address on an interface and enable it with **no shutdown**, 2 routes (per interface) will automatically be added to the routing table:
 → a **connected** route
 → a **local** route



Connected and Local routes

192 . 168 . 1 . 0 /24
 255 . 255 . 255 . 0

=FIXED (can't change)

=not fixed

```
C    192.168.1.0/24 is directly connected, GigabitEthernet0/2
```

- **192.168.1.0/24** matches 192.168.1.0 ~ 192.168.1.255.
→ If R1 receives a packet with a destination in that range, it will send the packet out of G0/2.

A route **matches** a packet's destination if the packet's destination IP address is part of the network specified in the route.

192.168.1.2 = match

→ Send packet out of G0/2

192.168.1.7 = match

→ Send packet out of G0/2

192.168.1.89 = match

→ Send packet out of G0/2

192.168.2.1 = no match

→ Send the packet using a different route, or drop the packet if there is no matching route.



Route Selection

```

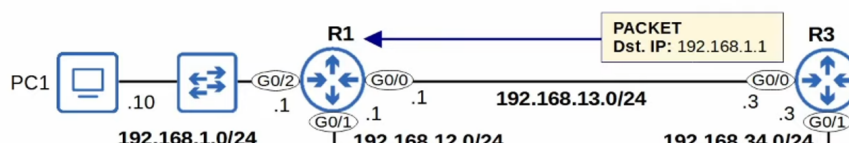
C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.1.0/24 is directly connected, GigabitEthernet0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet0/2
  
```

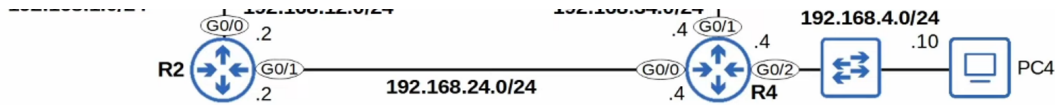
- A packet destined for **192.168.1.1** is matched by both routes:
192.168.1.0/24
192.168.1.1/32
- Which route will R1 use for a packet destined for 192.168.1.1?
 → It will choose the **most specific** matching route.
- The route to **192.168.1.0/24** includes 256 different IP addresses (192.168.1.0 – 192.168.1.255)
- The route to **192.168.1.1/32** includes only 1 IP address (192.168.1.1)
 → This route is more **specific**.
- **Most specific** matching route = the matching route with the **longest prefix length**.

When R1 receives a packet destined for 192.168.1.1, it will select the route to 192.168.1.1/32.

→ R1 will receive the packet for itself, rather than forward it out of G0/2.

Local route = keep the packet, don't forward





Term	Definition	Example
Connected Route (C)	A route to the network directly connected to one of the router's interfaces.	Interface G0/0 = 192.168.1.1/24 → Connected route = 192.168.1.0/24
Local Route (L)	A route to the specific IP address assigned to the router's own interface .	Interface G0/0 = 192.168.1.1/24 → Local route = 192.168.1.1/32



Summary

- Routers store information about destinations they know in their **routing table**.
→ When they receive packets, they look in the routing table to find the best route to forward the packet.
- Each **route** in the routing table is an instruction:
 - To reach destinations in network X, send the packet to **next-hop Y** (the next router in the path to the destination).
 - If the destination is directly connected (**Connected** route) send the packet directly to the destination.
 - If the destination is your own IP address (**Local** route), receive the packet for yourself.
 *We will look at how **next-hops** work in the next video on **static routes**.
- When you configure an IP address on an interface and enable the interface, two routes are automatically added to the routing table:
 - Connected** route (code **C** in the routing table): A route to the network connected to the interface.
→ ie. if the interface's IP is **192.168.1.1/24**, the route will be to **192.168.1.0/24**.
→ Tells the router: "To send a packet to a destination in this network, send it out of the interface specified in the route".
 - Local** route (code **L** in the routing table): A route to the exact IP address configured on the interface.
→ ie. if the interface's IP is **192.168.1.1/24**, the route will be to **192.168.1.1/32**.
→ Tells the router: "Packets to this destination are for you. You should receive them for yourself (not forward them)".
- A route **matches** a destination if the packet's destination IP address is part of the network specified in the route.
→ ie. a packet to **192.168.1.60** is matched by a route to **192.168.1.0/24**, but not by a route to **192.168.0.0/24**.
- If a router receives a packet and it doesn't have a route that matches the packet's destination, it will **drop** the packet.
→ This is different than switches, which **flood** frames if they don't have a MAC table entry for the destination.
- If a router receives a packet and it has multiple routes that match the packet's destination, it will use the **most specific matching route** to forward the packet.
 - **Most specific** matching route = the matching route with the longest prefix length.
 - This is different than switches, which look for an **exact** match in the MAC address table to forward frames.